

Spatial Behaviour Simulator

The problem

Buildings and public spaces are mostly designed using static plans, regulations, and assumptions. In practice, people often behave very differently than expected once the space is actually used.

This project could help prevent dangerous (or just annoying) crowding and improve evacuation planning in schools, stadiums, airports, and public events by simulating crowd behavior before real-world use.

Proposed Solution & Technical Approach

The proposed solution is an AI-powered spatial simulation platform where users can create a rough building layout and simulate how people move and behave inside it under different conditions.

The system focuses on movement, safety, and crowd behavior. AI agents react dynamically to obstacles, exits, crowd density, panic situations, and environmental conditions.

The platform could simulate:

- Normal crowd movement
- Emergency evacuations
- Congestion and bottlenecks
- Queue overflow situations
- Panic situations during large events

The software would generate analytics and visualizations such as crowd density heatmaps, evacuation estimates, and traffic flow analysis to identify potential design flaws before real-world use.

The technical stack for this project could look something like this (marked with * are technologies less commonly covered in Aalto courses):

Frontend

- React
- Three.js* for 3D visualization
- TailwindCSS

Backend

- Python backend (FastAPI or Flask)
- Agent-based simulation engine*
- WebSocket support for live updates*

AI / Simulation

Python-based AI agents
Pathfinding algorithms (A*)
Crowd simulation systems*
Behavioral models for movement prediction*

Data / Infrastructure

PostgreSQL
Docker

Target user group

Primary target users are:

Architects
Urban planners
Event organizers
Safety engineers

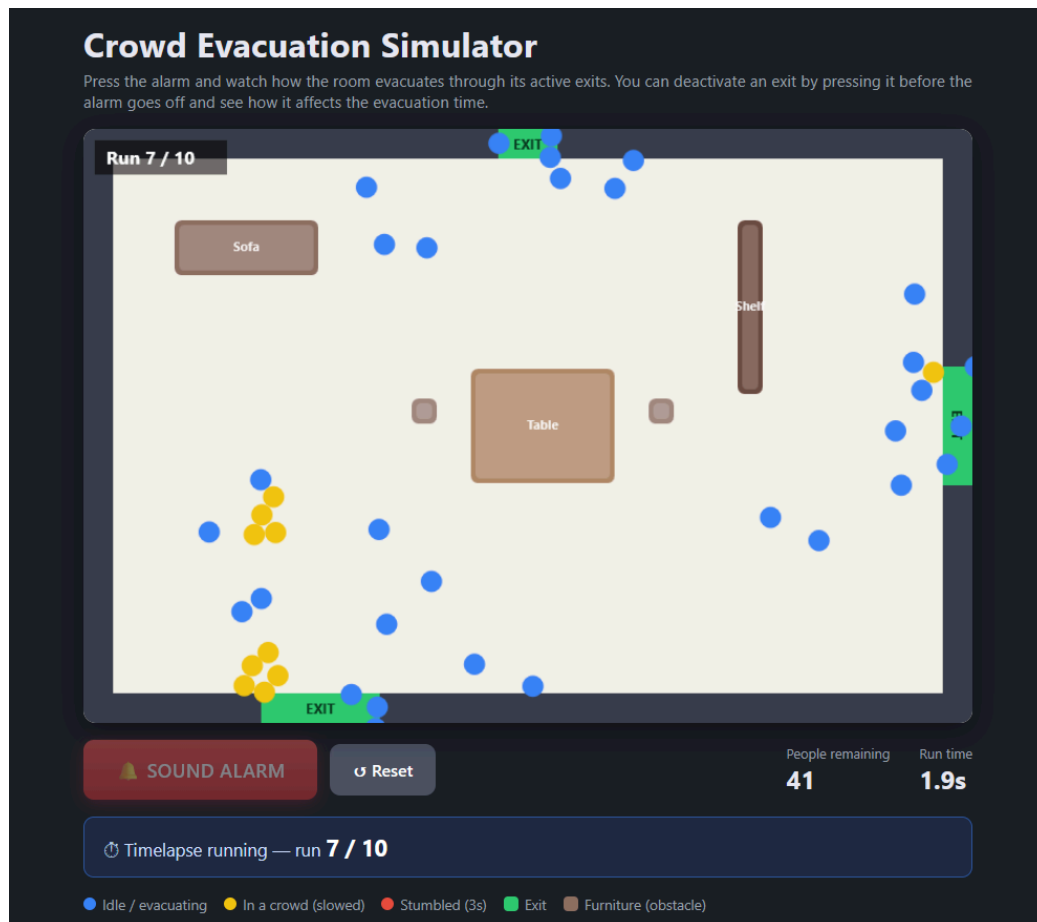
The platform is designed for organizations that need to safely manage large groups of people inside buildings or public spaces.

For example, architects could test evacuation routes and identify dangerous bottlenecks before construction. Schools and public institutions could simulate emergency situations, while event organizers could test how crowd movement changes based on entrance placement, queues, or restricted areas.

The tool could also support research and education in:

Architecture
AI systems
Human behavior simulation
Smart cities
Public safety

MVP



Above an image of the simulation running. It is a crowd evacuation simulator from a room with many 45 people initially and a couple of obstacles.

You can try it out at <https://caczyxsam.github.io/evacuation-sim/evacuation-sim.html>

This is a very simple example of a part of this quite large project. The complete project at least in our minds could have 3D spaces (as an option), AI that would give the people their functionality based on the context that the user inputs, and maybe even an option for AI to optimize the space for a certain purpose (which here would be getting everyone out) with given conditions like price, style, carrying capacity, etc.

Why is this an important issue?

This project has both practical and societal value. It could help architects and event organizers detect dangerous bottlenecks, improve evacuation routes, and optimize crowd flow before a building or event is used. This directly improves public safety in places such as schools, stadiums, airports, and concerts where poor crowd management can have serious consequences.